

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and executed system test plans for automotive camera and mechatronic ECU systems, ensuring comprehensive coverage against functional and performance requirements across the project lifecycle.
- Designed test specifications and methods for system-level testing of automotive software features, aligning test strategies with project schedules and maturity phases.
- Contributed to the review of system requirements and architecture designs, assessing testability and planning test coverage in collaboration with System Engineers and Function Owners.
- Implemented test automation capabilities following platform standards and project automation targets, enhancing efficiency in system validation processes.
- Supported trace analysis and troubleshooting to identify root causes of issues, validating fixes and maintaining detailed test results and logs throughout development.
- Collaborated with customer teams on issue reporting and joint testing events, contributing to reports and presentations on testing status and detected defects.

SKILLS

Testing & Validation – System Validation, Test Environment, Bench Testing, HIL Testing, Vehicle Testing, Test Automation, Stress Testing, Robustness Testing, Test Specification, Trace Analysis, Defect Reporting

Programming Languages – Python, C++, C, C#, .NET, NI LabVIEW, NI TestStand

Automotive Systems – ADAS, Mechatronic ECU Systems, Automotive Camera Systems, Software Features

Engineering Concepts – Electronic Systems, Systems Development Life Cycle, Requirements Engineering, Architecture Design

Tools & Methodologies – Git, Jenkins, Jira, Confluence, Agile, Scrum

PROJECTS

Automotive Camera System Validation Suite

- Developed a Python-based test automation suite for validating automotive camera systems, integrating with hardware-in-the-loop (HIL) test benches.
- Designed and implemented test cases covering functional requirements such as image processing, object detection, and performance metrics like frame rate stability.
- Managed test execution and defect tracking using Jira, providing detailed reports on test coverage and system maturity to the project team.

Mechatronic ECU Performance Testing Framework

- Engineered a test framework using NI LabVIEW and TestStand to execute performance and stress tests on mechatronic Electronic Control Units (ECUs).
- Developed scripts to simulate real-world driving scenarios and measure response times, power consumption, and error handling under load.
- Documented test procedures and results, contributing to the overall system validation report for a key automotive component.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)